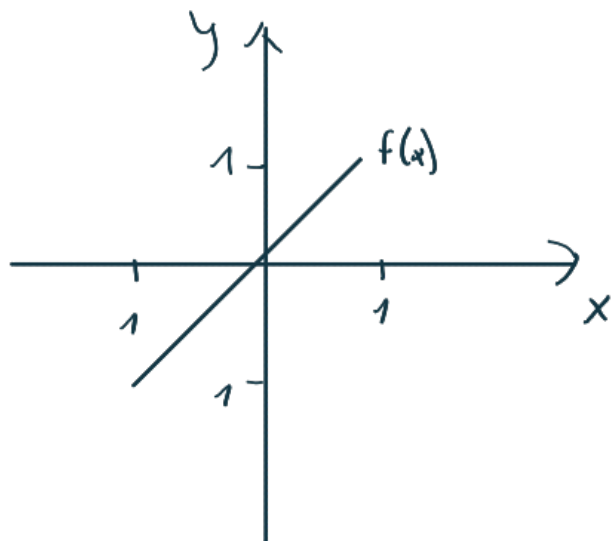


8. a) nariši:

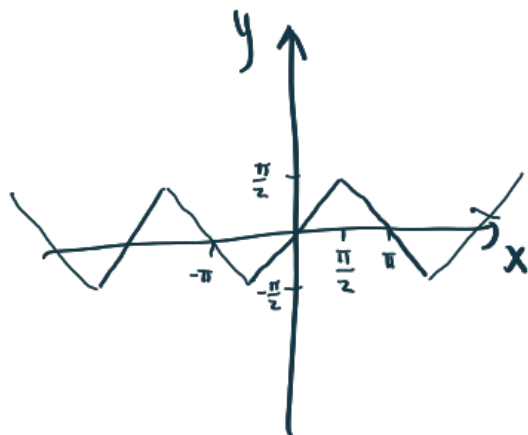
$$f(x) = \sin(\arcsin(x))$$



$$\sin: \mathbb{R} \rightarrow [-1, 1]$$

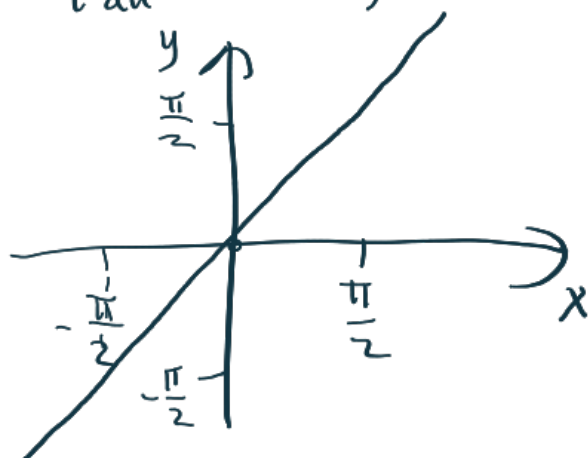
$$\arcsin: [-1, 1] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

$$g(x) = \arcsin(\sin(x))$$



b)

$$f(x) = \tan(\arctan(x))$$



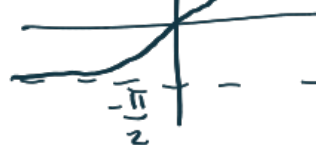
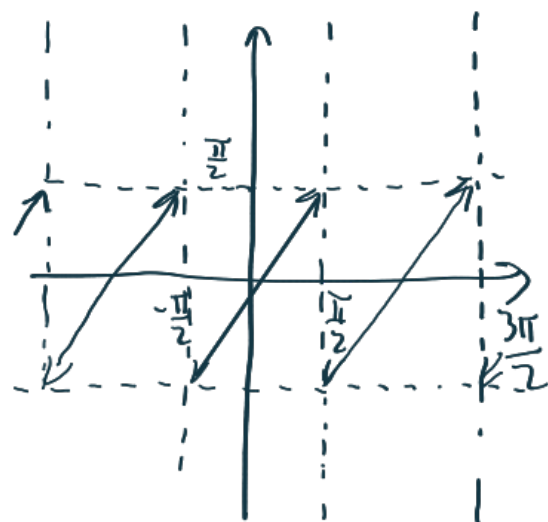
$$\tan: \mathbb{R} \setminus \left\{ \frac{\pi}{2} + k\pi; k \in \mathbb{Z} \right\} \rightarrow \mathbb{R}$$



$$\arctan: \mathbb{R} \rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

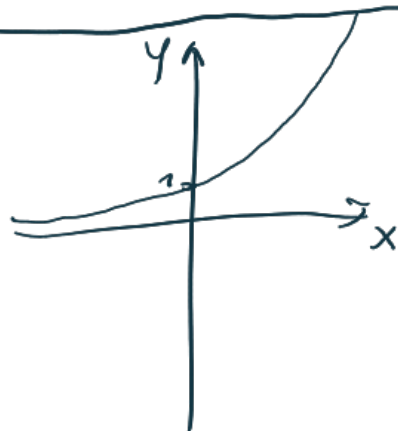
$$g(x) = \arctan(\tan(x))$$





9. levi / desni inverz

a) $f: \mathbb{R} \rightarrow \mathbb{R} \quad x \mapsto e^x$



je inj. $\Rightarrow$ ima l. inv.

ni surj. $\Rightarrow$ nima d. inv.

$$f^{-1}(x) = \ln x$$

L: $\ln(e^x) \checkmark$
D: $e^{\ln x}$ Ne obstaja, če $x < 0$

b) $f: \mathbb{R} \rightarrow [0, \infty) \quad x \mapsto x^2$

$$f^{-1}(x) = \sqrt{x}$$

ni inj. $\Rightarrow$ nima l. inv.

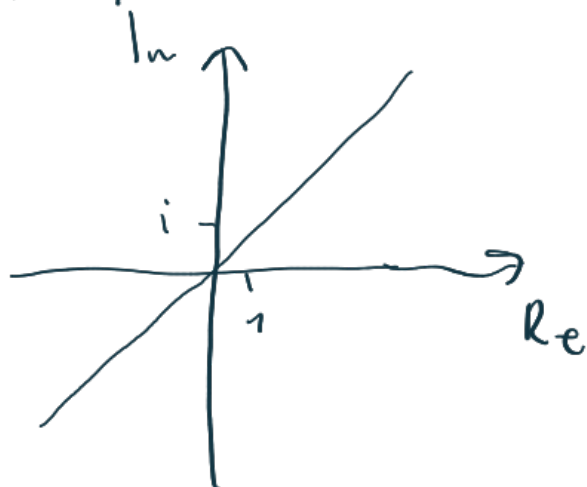
je surj. $\Rightarrow$ ima d. inv.

L: $\sqrt{x^2} \rightarrow 2$ mo žnosti za x
D: $(\sqrt{x})^2 \checkmark$
 $(\pm x) \Rightarrow$ ne obstaja

Števila

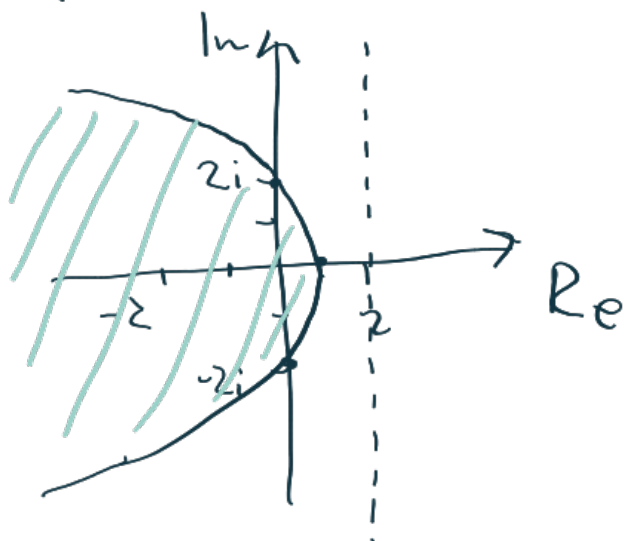
① Narišite in geon. interpr.

1) $|z-1| = |z-i|$



Razd. med 2 in 1 je enaka kot med z in i

2) $|z| + \operatorname{Re} z \leq 2$



$z = a + bi$

$\sqrt{a^2 + b^2} \leq 2 - a ; a \leq 2$

$a^2 + b^2 \leq 4 - 4a + a^2$

$b^2 - 4a \leq 4$

$b^2 \leq -2 \cdot 2(a-1)$

3) $|z-1| + |z+1| = 5$

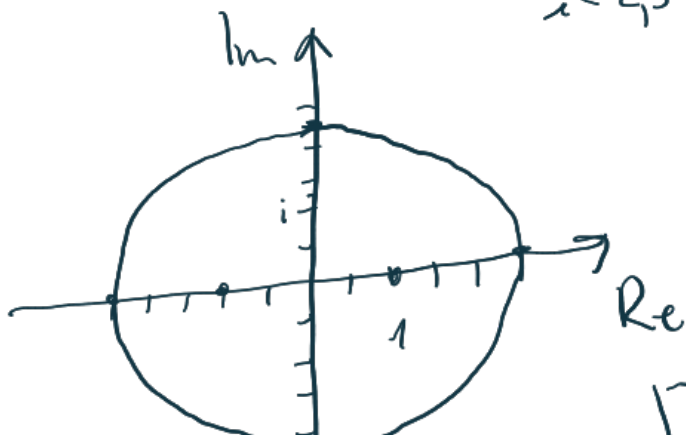
elipsa

$a = 2,5$



$b^2 = 2,5^2 - 1 = 5,25 = \frac{21}{4}$

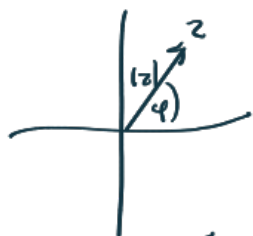
$b = \frac{\sqrt{21}}{2}$



$\sqrt{a^2 - b^2}$

$$\textcircled{2} \quad z = |z| (\cos \phi + i \sin \phi) =$$

$$= |z| e^{i\phi}$$



Izračunaj s pomočjo
formule:

$$\boxed{z^h = |z|^h e^{i h \phi}} = |z|^h (\cos h \phi + i \sin h \phi)$$

$$a) \quad z = (1 + i\sqrt{3})^{1/2}$$

$$|w| = \sqrt{1+3} = 2$$



$$\tan \phi = \sqrt{3}$$

$$\phi = \frac{\pi}{3}$$

$$1/2 : 3 = 1/6$$

$$z = w^{1/2} = |w|^{1/2} (\cos 1/2 \phi + i \sin 1/2 \phi) =$$

$$= 2^{1/2} (\cos 1/4 \pi + i \sin 1/4 \pi) = \underline{\underline{2^{1/2}}}$$

$$b) \quad z = \frac{1}{(1+i)^8} = \frac{1}{16 (\cos \frac{8\pi}{4} + i \sin \frac{8\pi}{4})} = \frac{1}{16}$$

